THE SHADOW OBSTRUCTION TOWER OF A CHIRAL ALGEBRA IS THE L_∞-FORMALITY OBSTRUCTION TOWER

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Abstract. Let $\mathcal{A}$ be a chiral algebra on a smooth projective curve and let $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ be its modular convolution dg Lie algebra, with bar-intrinsic Maurer–Cartan element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}})$. We prove that the shadow obstruction tower $(\mathrm{Sh}_r(\mathcal{A}))_{r\geq 2}$, defined as the finite-order projections of $\Theta_{\mathcal{A}}$ under the bar filtration, coincides with the L_∞-formality obstruction tower of the genus-zero part $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod},(0)}$. The identification is established at all degrees by induction from explicit base cases at $r = 2, 3, 4$. As a corollary the operadic complexity conjecture is proved: the shadow termination degree, the transferred A_∞-depth, and the L_∞-formality level of $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod},(0)}$ are a single invariant of the chiral algebra.

1. Introduction

1.1. **Formality and its obstructions.** A dg Lie algebra $\mathfrak{g}$ is L_∞-*formal* when its minimal model, obtained by homotopy transfer onto cohomology, carries trivial higher brackets $\ell_n = 0$ for $n \geq 3$; equivalently, $\mathfrak{g}$ is L_∞ quasi-isomorphic to $(H^*(\mathfrak{g}), 0, [-,-])$. Kontsevich's formality theorem [3] proved that polyvector fields on a smooth manifold form an L_∞-formal dg Lie algebra, giving deformation quantization of Poisson structures; Tamarkin [8] recovered the result operadically from formality of the little disks. When formality fails, obstructions are recorded in cohomology classes $o_3, o_4, \ldots$: at degree r the class o_r measures whether ℓ_r^{tr} vanishes on cohomology; if $o_r = 0$, a homotopy kills ℓ_r^{tr} and the next obstruction lives one degree higher. The sequence $(o_r)_{r\geq 3}$ is the *formality obstruction tower*; see Loday–Vallette [5]*Ch. 10.

1.2. **The shadow tower of a chiral algebra.** On a smooth projective curve X, a chiral algebra $\mathcal{A}$ [1] has a bar complex $\bar{B}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$, where $\bar{\mathcal{A}} = \ker(\epsilon)$ is the augmentation ideal, with deconcatenation coproduct and differential built from the Arnold form $\eta_{ij} = d\log(z_i - z_j)$ on the Fulton–MacPherson compactification of the configuration space. The modular convolution dg Lie algebra $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ is the pronilpotent dg Lie algebra whose differential encodes graph sums over stable curves in $\overline{\mathcal{M}}_{g,n}$. The bar-intrinsic

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element $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0 \in \mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}})$ satisfies $D\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$ and packages the full genus expansion of $\mathcal{A}$ [6]*Thm. H. Its finite-degree truncations $\Theta_{\mathcal{A}}^{\leq r}$ define the *shadow obstruction tower*

$$\mathrm{Sh}_r(\mathcal{A}) \;:=\; -\nabla_H^{-1}\big([\Theta_{\mathcal{A}}^{\leq r-1}, \Theta_{\mathcal{A}}^{\leq r-1}]_r^{(0)}\big), \qquad r \geq 2,$$

where the bracket is genus-zero graph composition and ∇_H is the Hessian covariant derivative on the shadow algebra $\mathcal{A}^{\mathrm{sh}}$. The first three shadows are $\mathrm{Sh}_2(\mathcal{A}) = \kappa(\mathcal{A})$ (the modular characteristic [6]*Thm. D), $\mathrm{Sh}_3(\mathcal{A}) = \mathfrak{C}(\mathcal{A})$ (cubic shadow), and $\mathrm{Sh}_4(\mathcal{A}) = \mathfrak{Q}(\mathcal{A})$ (quartic shadow).

1.3. **The main theorem.** The classical formality obstruction tower lives in Harrison cohomology of a dg Lie algebra; the shadow obstruction tower lives in the shadow algebra of a chiral algebra on a curve. Nothing in the definitions forces the two sequences to agree.

Theorem 1.1 (Shadow–formality identification; main theorem)**.** *For every chiral algebra $\mathcal{A}$ and every degree $r \geq 2$,*

$$\mathrm{Sh}_r(\mathcal{A}) \;=\; \ell_r^{(0),\mathrm{tr}}(\Theta_{\mathcal{A}}^{\leq r-1}, \ldots, \Theta_{\mathcal{A}}^{\leq r-1}),$$

where $\ell_r^{(0),\mathrm{tr}}$ is the transferred genus-zero L_∞ bracket of degree r on the minimal model of $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod},(0)}$. Consequently the shadow obstruction tower of $\mathcal{A}$ is identical, as a sequence of cohomology classes, to the L_∞-formality obstruction tower of $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod},(0)}$.

Theorem 1.1 is the self-contained extraction of the shadow–formality identification theorem in [6]*Ch. on higher-genus modular Koszul. The proof is by induction on r: the base cases $r = 2, 3, 4$ are verified explicitly (Proposition 4.1) and the inductive step uses the Kadeishvili–Merkulov tree formula [2, 7] on the strict genus-zero convolution algebra. The tree sum defining $\ell_r^{(0),\mathrm{tr}}$ and the graph sum defining Sh_r range over the same combinatorial data with matching propagators and coefficients.

1.4. **Consequences.** Theorem 1.1 resolves the operadic complexity conjecture: the shadow termination degree $r_{\max}(\mathcal{A})$, the transferred A_∞-depth $d_\infty(\mathcal{A})$, and the L_∞-formality level $f_\infty(\mathcal{A})$ of $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod},(0)}$ are a single invariant of $\mathcal{A}$ (Corollary **??**). The four shadow-depth classes are realized by Heisenberg (class **G**, $r_{\max} = 2$, formal), affine Kac–Moody (class **L**, $r_{\max} = 3$), $\beta\gamma$ (class **C**, $r_{\max} = 4$), and Virasoro (class **M**, $r_{\max} = \infty$, intrinsically non-formal); see Section 7.

1.5. **Organization.** Section 2 defines the shadow obstruction tower. Section 3 recalls the transferred L_∞ structure and the formality obstruction tower. Section 4 proves the identification at degrees $2, 3, 4$. Section 5 proves Theorem 1.1. Section 6 deduces the operadic complexity corollary. Section 7 verifies the four standard families.

Conventions. Grading is cohomological with $|d| = +1$; desuspension lowers degree, $|s^{-1}v| = |v| - 1$. Bar complexes use the augmentation ideal: $\bar{B}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$. The propagator $d\log E(z,w)$ on the bar complex has weight one. All statements are at the level of cohomology classes unless otherwise noted.

2. The shadow obstruction tower

2.1. **The bar complex and its convolution algebra.** Let X be a smooth projective curve over $\mathbb{C}$ and let $\mathcal{A}$ be a chiral algebra on X [1]. The ordered chiral bar complex $\bar{B}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}}) = \bigoplus_{n\geq 0}(s^{-1}\bar{\mathcal{A}})^{\otimes n}$ carries the deconcatenation coproduct, with differential assembled from $\eta_{ij} = d\log(z_i - z_j)$ and the chiral product $\mu\colon j_*j^*(\mathcal{A}\boxtimes\mathcal{A}) \to \Delta_*\mathcal{A}$. The *modular convolution dg Lie algebra* is

$$\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}} \;:=\; \mathrm{Hom}^{\mathrm{cyc}}\big(\bar{B}(\mathcal{A}),\, \mathcal{A}\big) \otimes \mathsf{G},$$

where $\mathsf{G} = \bigoplus_{g\geq 0}\mathsf{G}^{(g)}$ is the coefficient algebra generated by stable graphs on $\overline{\mathcal{M}}_{g,n}$. The genus filtration gives $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}} = \bigoplus_{g\geq 0}\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}{}_g$; the genus-zero part $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod},(0)} = \mathrm{Hom}^{\mathrm{cyc}}(\bar{B}(\mathcal{A}),\mathcal{A}) \otimes \mathsf{G}^{(0)}$ is a strict dg Lie algebra with bracket $[-,-]^{(0)}$ from tree composition, since genus-zero stable graphs are trees.

2.2. **The Maurer–Cartan element and degree truncations.** The differential on $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ splits as $D = d_0 + D_{\mathrm{int}}$, with d_0 linear and D_{int} the interacting piece encoding the chiral product and the Arnold form. The difference $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0 \in \mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ is a Maurer–Cartan element [6]*Thm. on MC2 bar-intrinsic:

$$D\Theta_{\mathcal{A}} + \tfrac{1}{2}[\Theta_{\mathcal{A}},\Theta_{\mathcal{A}}] = 0. \tag{2.1}$$

The element $\Theta_{\mathcal{A}}$ is bar-intrinsic. The bar complex is filtered by tensor degree, $F^r\bar{B}(\mathcal{A}) = \bigoplus_{n\leq r}(s^{-1}\bar{\mathcal{A}})^{\otimes n}$, and the truncations $\Theta_{\mathcal{A}}^{\leq r} := \Theta_{\mathcal{A}} \bmod F^{r+1}$ form a tower in which each term is determined up to a coboundary by the previous one, via the Maurer–Cartan recursion.

Definition 2.1 (Shadow obstruction tower)**.** The *degree-r shadow* of $\mathcal{A}$ is

$$\mathrm{Sh}_r(\mathcal{A}) \;:=\; -\nabla_H^{-1}\big([\Theta_{\mathcal{A}}^{\leq r-1},\Theta_{\mathcal{A}}^{\leq r-1}]_r^{(0)}\big) \;\in\; \mathcal{A}_{r,0}^{\mathrm{sh}}, \tag{2.2}$$

where the bracket is the genus-zero graph composition, the subscript r extracts the degree-r component, and ∇_H is the Hessian covariant derivative on the shadow algebra $\mathcal{A}^{\mathrm{sh}} = H^*(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod},(0)})$ descended from the convolution Lie bracket. The sequence $(\mathrm{Sh}_r(\mathcal{A}))_{r\geq 2}$ is the *shadow obstruction tower* of $\mathcal{A}$.

The definition is bar-intrinsic: if $\Theta_{\mathcal{A}} \equiv \Theta'_{\mathcal{A}}$ mod gauge, then $\mathrm{Sh}_r(\mathcal{A})$ and $\mathrm{Sh}'_r(\mathcal{A})$ coincide as cohomology classes in $\mathcal{A}_{r,0}^{\mathrm{sh}}$.

Remark 2.2 (Closed formulas). On the scalar lane, writing $\Theta_{\mathcal{A}} = \kappa\cdot\eta\otimes\Lambda$, one has $\mathrm{Sh}_2 = \kappa$, $\mathrm{Sh}_3 = \mathfrak{C}$, $\mathrm{Sh}_4 = \mathfrak{Q}$, and the shadow metric identity $H(t)^2 = 4\kappa^2 + 12\kappa\mathfrak{C}\,t + (9\mathfrak{C}^2 + 16\kappa\mathfrak{Q})\,t^2 + \cdots$ determines Sh_r for $r \geq 5$ from $(\kappa,\mathfrak{C},\mathfrak{Q})$.

3. The L_∞-formality obstruction tower

3.1. **Homotopy transfer on a strict model.** Let $\mathfrak{g}$ be a strict dg Lie algebra with differential d and bracket $[-,-]$. A *deformation retract* onto $H^*(\mathfrak{g})$ is a triple (ι,π,h) with $\iota\colon H^*(\mathfrak{g})\to\mathfrak{g}$, $\pi\colon\mathfrak{g}\to H^*(\mathfrak{g})$, and $h\colon\mathfrak{g}\to\mathfrak{g}$ of degree -1, satisfying $\pi\iota=\mathrm{id}$ and $\mathrm{id}-\iota\pi=[d,h]$. The homotopy transfer theorem of Kadeishvili [2] and Kontsevich–Soibelman [4], in the form of Loday–Vallette [5]*§10.3, produces a transferred L_∞-algebra structure $(H^*(\mathfrak{g}),0,\{\ell_n^{\mathrm{tr}}\}_{n\geq 2})$ quasi-isomorphic to $\mathfrak{g}$, with brackets computed by the Kadeishvili–Merkulov tree sum

$$\ell_r^{\mathrm{tr}}(a_1,\dots,a_r)\;=\;\sum_{T\in\mathsf{Trees}_r}\frac{\pm 1}{|\mathrm{Aut}(T)|}\,\pi\circ\Phi_T(\iota(a_1),\dots,\iota(a_r)),\tag{3.1}$$

where Trees_r is the set of planar rooted trees with r leaves and Φ_T decorates internal vertices with $[-,-]$ and internal edges with h.

3.2. **The formality obstruction tower.**

Definition 3.1 (L_∞-formality and obstructions)**.** The dg Lie algebra $\mathfrak{g}$ is *L_∞-formal* if there is an L_∞ quasi-isomorphism $\mathfrak{g}\simeq(H^*(\mathfrak{g}),0,[-,-])$; equivalently, $\ell_n^{\mathrm{tr}}=0$ in L_∞ cohomology for all $n\geq 3$. The *L_∞-formality obstruction tower* is the sequence

$$\big(o_r(\mathfrak{g})\big)_{r\geq 3},\qquad o_r(\mathfrak{g})\;:=\;[\ell_r^{\mathrm{tr}}]\;\in\;H^*\big(\mathrm{Harr}^\bullet(\mathfrak{g})\big).$$

If $o_r(\mathfrak{g})=0$, an L_∞ homotopy kills ℓ_r^{tr} and the next obstruction is defined.

For a pronilpotent dg Lie algebra with Maurer–Cartan element Θ, evaluating the transferred brackets on Θ produces a refined tower $(\ell_r^{\mathrm{tr}}(\Theta,\dots,\Theta))_{r\geq 2}$, which takes values in the convolution target rather than in Harrison cohomology; this refined tower is the one compared with the shadow tower.

Remark 3.2 (Harrison equals cyclic at genus zero)*.* For $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod},(0)}$, the Harrison complex controlling L_∞-formality coincides with the genus-zero cyclic deformation complex $\mathrm{Def}_{\mathrm{cyc}}^{(0)}(\mathcal{A})$, since genus-zero stable graphs are trees and the Harrison differential on trees is precisely operadic composition.

4. Base cases: degrees 2, 3, 4

Proposition 4.1 (Shadow–formality identification at low degree)**.** *For every chiral algebra $\mathcal{A}$ the following identities hold:*

$$\begin{aligned}
&(i)\ \mathrm{Sh}_2(\mathcal{A})\;=\;\kappa(\mathcal{A})\;=\;\ell_2^{(0)}(\Theta_{\mathcal{A}},\Theta_{\mathcal{A}}).\\
&(ii)\ \mathrm{Sh}_3(\mathcal{A})\;=\;\mathfrak{C}(\mathcal{A})\;=\;-h\big(\ell_3^{(0)}(\Theta_{\mathcal{A}}^{\leq 2},\Theta_{\mathcal{A}}^{\leq 2},\Theta_{\mathcal{A}}^{\leq 2})\big).\\
&(iii)\ \mathrm{Sh}_4(\mathcal{A})\;=\;\mathfrak{Q}(\mathcal{A})\;=\;\big[\ell_4^{(0),\mathrm{tr}}(\Theta_{\mathcal{A}}^{\leq 3},\Theta_{\mathcal{A}}^{\leq 3},\Theta_{\mathcal{A}}^{\leq 3},\Theta_{\mathcal{A}}^{\leq 3})\big].
\end{aligned}$$

Proof. Part (i). The binary bracket on $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod},(0)}$ is the genus-zero convolution Lie bracket; evaluating on $\Theta_{\mathcal{A}}$ produces the scalar curvature $\kappa(\mathcal{A})\in\mathbb{Q}(c)$ by definition [6]*Thm. D. The identity $\mathrm{Sh}_2(\mathcal{A})=\kappa(\mathcal{A})$ is the degree-2 shadow on the scalar lane, so the three quantities agree.

Part (ii). The ternary bracket $\ell_3^{(0)}$, computed by (3.1), is a sum over $\mathsf{Trees}_3 = \{T_s, T_t, T_u\}$, the three binary trees with three leaves. On the primary line $[\ell_3^{(0)}] = 0$ since $H^1(\overline{\mathcal{M}}_{0,4}) = H^1(\mathbb{P}^1) = 0$; at the chain level $\ell_3^{(0)}$ is exact and nonzero, and homotopy transfer gives $\mathfrak{C}(\mathcal{A}) = -h(\ell_3^{(0)}(\Theta_{\mathcal{A}}^{\leq 2}, \Theta_{\mathcal{A}}^{\leq 2}, \Theta_{\mathcal{A}}^{\leq 2}))$. The cubic shadow $\mathfrak{C}(\mathcal{A})$, computed as the three-channel graph sum at degree three, matches this three-tree sum because each binary tree with three leaves corresponds to one of the s-, t-, u-channel genus-zero stable graphs with four external legs, with identical weights and propagators.

Part (iii). The quaternary bracket $\ell_4^{(0)}$ is a sum over the 15 trivalent trees with five leaves, the faces of the Stasheff associahedron K_4. Its transferred class $\ell_4^{(0),\mathrm{tr}}$ lies in $H^*(\overline{\mathcal{M}}_{0,5})$. The weight-four graph sum defining $\mathfrak{Q}(\mathcal{A})$ enumerates precisely the same 15 trees with matching weights, signs, and propagators [6]*App. on nonlinear modular shadows, so $\mathrm{Sh}_4(\mathcal{A}) = \mathfrak{Q}(\mathcal{A}) = [\ell_4^{(0),\mathrm{tr}}(\Theta_{\mathcal{A}}^{\leq 3}, \ldots, \Theta_{\mathcal{A}}^{\leq 3})]$. □

Remark 4.2 (Degree-2: κ is family-dependent). The identity $\mathrm{Sh}_2(\mathcal{A}) = \kappa(\mathcal{A})$ holds at degree two on the scalar lane for every chiral algebra in the standard landscape; on the primary line this is the definition $S_2 := \kappa$. What is family-dependent is the value of κ itself: $\kappa^{\mathrm{Vir}} = c/2$, $\kappa^{\mathrm{Heis}} = k$, $\kappa^{\mathrm{KM}} = \dim(\mathfrak{g})(k + h^\vee)/(2h^\vee)$, $\kappa^{\mathcal{W}_N} = c(H_N - 1)$ with $H_N = \sum_{j=1}^N 1/j$. The coincidence $\kappa = c/2$ holds only for Virasoro, not universally; writing $\kappa = c/2$ for a non-Virasoro family is a standard error.

5. The main theorem

Theorem 5.1 (Shadow obstruction tower equals formality obstruction tower)**.** *For every chiral algebra $\mathcal{A}$ and every degree $r \geq 2$,*

$$\mathrm{Sh}_r(\mathcal{A}) \;=\; \ell_r^{(0),\mathrm{tr}}(\Theta_{\mathcal{A}}^{\leq r-1}, \ldots, \Theta_{\mathcal{A}}^{\leq r-1}), \tag{5.1}$$

where $\ell_r^{(0),\mathrm{tr}}$ is the transferred genus-zero L_∞ bracket on the minimal model of $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod},(0)}$. Consequently the shadow obstruction tower of $\mathcal{A}$ coincides with the L_∞-formality obstruction tower of $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod},(0)}$ evaluated on the truncated Maurer–Cartan element.

Proof. Induction on the degree r. The base cases $r = 2, 3, 4$ are Proposition 4.1. The inductive step has four parts.

Step 1: Restriction to genus zero. Since genus-zero stable graphs are trees, $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod},(0)}$ is a strict dg Lie algebra and the Harrison complex controlling L_∞-formality coincides with the genus-zero cyclic deformation complex $\mathrm{Def}_{\mathrm{cyc}}^{(0)}(\mathcal{A})$. The Harrison differential on trees is precisely operadic composition on $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod},(0)}$, so $\mathrm{Harr}^\bullet(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod},(0)}) \xrightarrow{\sim} \mathcal{A}_{*,0}^{\mathrm{sh}}$ is an isomorphism, not merely an injection.

Step 2: Homotopy transfer. Choose a deformation retract $(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod},(0)}, h, \mathcal{A}_{*,0}^{\mathrm{sh}})$ with $\pi\iota = \mathrm{id}$, $\mathrm{id} - \iota\pi = [D, h]$. The homotopy transfer theorem [2, 4, 5]

produces a transferred L_∞ structure $(\mathcal{A}^{\mathrm{sh}}_{*,0}, \{\ell_r^{(0),\mathrm{tr}}\})$, with brackets computed by (3.1).

Step 3: Tree formula equals shadow graph sum. The degree-r shadow $\mathrm{Sh}_r(\mathcal{A})$, by (2.2), expands $[\Theta_{\mathcal{A}}^{\leq r-1}, \Theta_{\mathcal{A}}^{\leq r-1}]_r^{(0)}$ as a sum over genus-zero stable graphs with r external legs, with internal vertices decorated by lower-degree shadows $\mathrm{Sh}_j(\mathcal{A})$ $(j < r)$ and internal edges by the propagator $P_{\mathcal{A}} = \nabla_H^{-1}$. These are the same trees and the same propagators as in (3.1): (a) a genus-zero stable graph with r external legs is a planar rooted tree $T \in \mathsf{Trees}_r$ with valences determined by the graph; (b) the shadow propagator $P_{\mathcal{A}} = h \circ \pi$ corresponds to the homotopy h; (c) the vertex evaluation $\mathrm{Sh}_j = \pi \circ \ell_j^{(0)} \circ \iota^{\otimes j}$ corresponds to the vertex decoration $[-,-]^{(0)}$.

Step 4: Induction. At $r = 2$, both sides equal $\kappa(\mathcal{A})$ by Proposition 4.1(i); the cases $r = 3, 4$ are parts (ii), (iii) of the same proposition. Assuming (5.1) through degree $r-1$, the graph sum for $\mathrm{Sh}_r(\mathcal{A})$ and the tree formula for $\ell_r^{(0),\mathrm{tr}}(\Theta^{\leq r-1}, \dots, \Theta^{\leq r-1})$ range over identical planar rooted trees with identical internal-vertex decorations (via the inductive hypothesis), propagators, and signs. The two sums therefore agree as chains in $\mathcal{A}^{\mathrm{sh}}_{*,0}$, completing the induction. □

Remark 5.2 (Genus corrections decouple; uniform-weight scope). Theorem 5.1 is a genus-zero statement. On the uniform-weight scalar lane, the shadow obstruction tower $\mathrm{Sh}_r(\mathcal{A})$ decomposes, at each genus $g \geq 0$ separately, as

$$\mathrm{Sh}_r(\mathcal{A}) \;=\; \underbrace{\ell_r^{(0),\mathrm{tr}}(\Theta_{\mathcal{A}}^{\leq r-1}, \dots)}_{\text{genus-zero formality}} + \underbrace{\sum_{g\geq 1} \ell_r^{(g),\mathrm{tr}}(\dots) + \Lambda_P(\dots)}_{\text{genus } \geq 1 \text{ corrections}},$$

where Λ_P is the BV loop operator. The classical and quantum contributions decouple at the level of the shadow decomposition: the quantum corrections form a parallel tower controlled by the quantum L_∞ structure, not by classical formality theory. At genus $g \geq 2$ with multi-weight generators, the scalar formula for the genus corrections receives a cross-channel term $\delta F_g^{\mathrm{cross}}$; the genus-zero identification (5.1) nevertheless holds verbatim.

6. Operadic complexity, proved

Definition 6.1 (Three complexity invariants)**.** For a chiral algebra $\mathcal{A}$, set $r_{\max}(\mathcal{A}) := \sup\{r \geq 2 : \mathcal{A}^{\mathrm{sh}}_{r,0} \neq 0\}$ (shadow termination degree), $d_\infty(\mathcal{A}) := \sup\{n \geq 2 : m_n^{\mathrm{tr}} \neq 0\}$ (transferred A_∞-depth on the minimal model of $\mathrm{Def}_{\mathrm{cyc}}(\mathcal{A})$), and $f_\infty(\mathcal{A}) := \sup\{n \geq 2 : \ell_n^{(0),\mathrm{tr}} \neq 0\}$ (L_∞-formality level of $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod},(0)}$).

Corollary 6.2 (Operadic complexity)**.** $r_{\max}(\mathcal{A}) = d_\infty(\mathcal{A}) = f_\infty(\mathcal{A})$ *for every chiral algebra* $\mathcal{A}$.

Proof. The equality $r_{\max} = f_\infty$ is immediate from Theorem 5.1: $\mathrm{Sh}_r(\mathcal{A}) = 0$ iff $\ell_r^{(0),\mathrm{tr}}$ vanishes on the truncated Maurer–Cartan element. For $f_\infty = d_\infty$,

the tree formula gives $\ell_r^{(0),\mathrm{tr}} = \mathrm{Alt}(m_r^{\mathrm{tr}})$; since antisymmetrization is injective on cyclic cochains, $m_r^{\mathrm{tr}} = 0$ iff $\ell_r^{(0),\mathrm{tr}} = 0$. □

Corollary 6.3 (Depth classification)**.** *The standard landscape of chiral algebras is partitioned by the shadow depth into four classes, each realized by a standard family:*

$$\boldsymbol{G}\ (r_{\max} = 2), \quad \boldsymbol{L}\ (r_{\max} = 3), \quad \boldsymbol{C}\ (r_{\max} = 4), \quad \boldsymbol{M}\ (r_{\max} = \infty).$$

The class $\boldsymbol{G}$ *consists of* L_∞*-formal algebras; the class* $\boldsymbol{M}$ *consists of intrinsically non-formal algebras, in the sense that no finite truncation of their minimal* L_∞ *model is quasi-isomorphic to a dg Lie algebra.*

Proof. Theorem 5.1 identifies shadow depth with L_∞-formality level, and Corollary 6.2 identifies both with the transferred A_∞-depth. The depth is therefore an invariant of the L_∞ quasi-isomorphism class of $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod},(0)}$. Examples in Section 7 realize each class. □

7. Examples

The four standard families realize the four shadow-depth classes.

Example 7.1 (Heisenberg; class **G**, formal)**.** The Heisenberg vertex algebra Heis_k at level k, with $J(z)J(w) \sim k/(z-w)^2$, has $\kappa(\mathrm{Heis}_k) = k$. The bar differential is strictly commutative, so $m_n = 0$ and $\ell_n^{(0),\mathrm{tr}} = 0$ for $n \geq 3$. Hence $r_{\max}(\mathrm{Heis}_k) = 2$ and Heis_k is L_∞-formal. Class **G**.

Example 7.2 (Affine Kac–Moody; class **L**)**.** The affine Kac–Moody vertex algebra $\widehat{\mathfrak{g}}_k$ has $\kappa(\widehat{\mathfrak{g}}_k) = \dim(\mathfrak{g})(k+h^\vee)/(2h^\vee)$. The binary bracket is the Lie bracket on $\mathfrak{g}$; Jacobi kills all degree-four graph contributions, so $\ell_4^{(0),\mathrm{tr}} = 0$, while $\ell_3^{(0),\mathrm{tr}} \neq 0$ in general. Hence $r_{\max}(\widehat{\mathfrak{g}}_k) = 3$. Class **L**.

Example 7.3 ($\beta\gamma$; class **C**, shadow depth 4)**.** The rank-one bosonic ghost system $\beta\gamma$ has $r_{\max}(\beta\gamma) = 4$: the brackets $\ell_3^{(0),\mathrm{tr}}$ and $\ell_4^{(0),\mathrm{tr}}$ are nonzero, but rank-one rigidity kills degree-five graphs, giving $\ell_n^{(0),\mathrm{tr}} = 0$ for $n \geq 5$. The primary invariants $(p_{\max}, k_{\max}, r_{\max}) = (1, 0, 4)$ are pairwise distinct: simple OPE pole, zero collision depth, shadow depth four. Class **C**.

Example 7.4 (Virasoro; class **M**, obstructed at every degree)**.** The Virasoro vertex algebra Vir_c has $\kappa(\mathrm{Vir}_c) = c/2$. The transferred brackets on the primary line of $H^*(\bar{B}(\mathrm{Vir}_c))$ satisfy $m_k^{\mathrm{tr}}(s^{-1}T, \ldots, s^{-1}T) = S_k \cdot e_{2k}$ for $k \geq 2$, with $s^{-1}T$ the desuspension of the conformal vector to bar degree one and e_{2k} the weight-$2k$ basis vector of Ext^1. The first shadow coefficients are

$$S_2 = \tfrac{c}{2}, \qquad S_3 = 2\ (c\text{-independent}), \qquad S_4 = \tfrac{10}{c(5c+22)},$$
$$S_5 = -\tfrac{48}{c^2(5c+22)}, \qquad S_6 = \tfrac{80(45c+193)}{3c^3(5c+22)^2}, \qquad S_7 = -\tfrac{2880(15c+61)}{7c^4(5c+22)^2}.$$

All S_k are nonzero for generic c, so $r_{\max}(\mathrm{Vir}_c) = \infty$: Virasoro is intrinsically non-formal. The Maurer–Cartan recursion forces every coefficient to be

nonzero, since the normal-ordered product $:T^n:$ generates operations at every degree. Class $\mathbf{M}$.

Remark 7.5 (The four classes exhaust the standard landscape). Classes $\mathbf{G}, \mathbf{L}, \mathbf{C}, \mathbf{M}$ are the only possibilities consistent with the shadow metric identity $H(t)^2 = t^4 Q(t)$ with $\deg Q \leq 2$. A finite depth in $\{5, 6, \dots\}$ would force a higher-degree Q, excluded by the algebraicity theorem for the shadow generating function [6]*Thm. on Riccati algebraicity. Depth is therefore 2, 3, 4, or ∞, and each is realized.

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